

Assignment: DHCP Server Configuration in Windows Server

1. Create a DHCP Scope Named "CampusWiFi"

Steps:

1. Open **Server Manager**.
2. Go to **Tools → DHCP**.
3. Expand the **Server Name**.
4. Expand **IPv4**.
5. Right-click **IPv4** and select **New Scope**.
6. Click **Next**.
7. Enter the Scope Name as **CampusWiFi**.
8. Configure the following IP address range:

Start IP Address: 192.168.50.100

End IP Address: 192.168.50.200

Subnet Mask: 255.255.255.0

9. Click **Next**.
10. Leave the exclusions empty unless specific exclusions are required.
11. Continue with the default settings.
12. Select **Yes, I want to activate this scope now**.
13. Click **Finish**.

Result:

A DHCP scope named **CampusWiFi** is successfully created with the IP address range:

192.168.50.100 – 192.168.50.200

Subnet Mask: 255.255.255.0

2. Create a DHCP Reservation

Steps:

1. In the DHCP Console, expand:

IPv4 → CampusWiFi → Reservations

2. Right-click **Reservations** and select **New Reservation**.
3. Enter the following details:

Reservation Name: Laptop

IP Address: 192.168.50.150

MAC Address: Enter your laptop's MAC address

Example: 00-15-5D-01-23-45

Description: Reserved IP for Laptop

4. Under Supported Types, select **Both**.
5. Click **Add**.
6. Click **Close**.

Result:

The specified device is reserved to receive the IP address:

192.168.50.150

from the **CampusWiFi** DHCP scope.

3. Change the DHCP Lease Duration to 2 Days

Steps:

1. Right-click the **CampusWiFi** scope.
2. Select **Properties**.
3. Open the **General** tab.
4. Under **Lease Duration for DHCP Clients**, change the values to:

Days: 2

Hours: 0

Minutes: 0

5. Click **Apply**.
6. Click **OK**.

Result:

The DHCP lease duration is successfully changed to **2 days**.

4. Verify That the Client Receives an IP Address

Steps:

1. Connect the client computer or virtual machine to the same network as the DHCP server.
2. Open **Command Prompt**.
3. Run the following commands:

```
ipconfig /release
```

```
ipconfig /renew
```

```
ipconfig
```

4. Verify that the client receives an IP address within the range:

192.168.50.100 – 192.168.50.200

5. If the reserved device is being used, verify that it receives:

192.168.50.150

6. Take a screenshot of the **ipconfig** output or **Network Settings** showing the assigned IP address.

Result:

The client successfully receives an IP address from the **CampusWiFi DHCP scope**.

Screenshot:

Insert a screenshot showing the assigned IP address.

5. Documentation of the DHCP Setup

The following details should be included in the Word or PDF report.

Title:

DHCP Server Configuration – CampusWiFi

Contents:

1. Installation of the DHCP Server Role

The DHCP Server role was installed successfully on Windows Server.

2. Creation of the CampusWiFi Scope

A DHCP scope named **CampusWiFi** was created successfully.

3. IP Address Range

Start IP Address: 192.168.50.100

End IP Address: 192.168.50.200

4. Subnet Mask

255.255.255.0

5. DHCP Reservation

Reservation Name: Laptop

Reserved IP Address: 192.168.50.150

Device MAC Address: Enter your laptop's MAC address

6. Lease Duration

The DHCP lease duration was changed to:

2 Days, 0 Hours, 0 Minutes

7. Client IP Address Verification

The client successfully obtained an IP address from the **CampusWiFi** DHCP scope.

8. Required Screenshots

The following screenshots should be included in the report:

1. DHCP Scope Creation
 2. DHCP Reservation Configuration
 3. Lease Duration Settings
 4. Client IP Address using **ipconfig**
 5. DHCP Scope showing the configured IP address range
-

```
Administrator: C:\Windows\system32\cmd.exe
C:\Users\Administrator>ipconfig /release

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    Default Gateway . . . . . : 

C:\Users\Administrator>ipconfig /renew

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : server.com
    IPv4 Address. . . . . : 192.168.1.50
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

C:\Users\Administrator>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : client1
    Primary Dns Suffix . . . . . : server.com
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
    DNS Suffix Search List. . . . . : server.com

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : server.com
    Description . . . . . : Intel(R) 82574L Gigabit Network Connection
    Physical Address. . . . . : 00-0C-29-C0-E1-A3
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    IPv4 Address. . . . . : 192.168.1.50(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : 29 June 2026 14:28:39
    Lease Expires . . . . . : 30 June 2026 14:28:38
```

File Action View Help			
DHCP	Start IP Address	End IP Address	Description
server.server.co	192.168.1.50	192.168.1.100	Address range for distribution
IPv4			
Server			
Scope			
Adc			
Adc			
Res			
Sco			
Pol			
Policies			
Filters			
IPv6			